

Symmetry Method Homework 7

1). From Gell-Mann matrices, $\hat{\lambda}_i$, $i=1, \dots, 8$
the generators are defined as $\hat{F}_i = \frac{\hat{\lambda}_i}{2}$

Show that

$$1.1) [\hat{F}_1, \hat{F}_2] = i\hat{F}_3 \quad 1.2) [\hat{F}_1, \hat{F}_4] = i\left(\frac{1}{2}\right)\hat{F}_7$$

$$1.3) [\hat{F}_1, \hat{F}_5] = i\left(-\frac{1}{2}\right)\hat{F}_6 \quad 1.4) [\hat{F}_2, \hat{F}_4] = i\left(\frac{1}{2}\right)\hat{F}_6$$

$$1.5) [\hat{F}_2, \hat{F}_5] = i\left(\frac{1}{2}\right)\hat{F}_7 \quad 1.6) [\hat{F}_3, \hat{F}_4] = i\left(\frac{1}{2}\right)\hat{F}_5$$

$$1.7) [\hat{F}_3, \hat{F}_6] = i\left(-\frac{1}{2}\right)\hat{F}_7 \quad 1.8) [\hat{F}_4, \hat{F}_5] = i\left(\frac{\sqrt{3}}{2}\right)\hat{F}_8$$

$$1.9) [\hat{F}_6, \hat{F}_7] = i\left(\frac{\sqrt{3}}{2}\right)\hat{F}_8$$

2). From the Cartan-Weyl basis:

$$\hat{T}_{\pm} = \hat{F}_1 \pm i\hat{F}_2, \quad \hat{V}_{\pm} = \hat{F}_4 \pm i\hat{F}_5, \quad \hat{U}_{\pm} = \hat{F}_6 \pm i\hat{F}_7$$

$$\hat{T}_3 = \hat{F}_3, \quad \hat{Y} = \frac{2}{\sqrt{3}}\hat{F}_8.$$

2.1) In the adjoint representation, find

the weight vector for

$$\{|\hat{T}_3\rangle, |\hat{Y}\rangle, |\hat{T}_+\rangle, |\hat{T}_-\rangle, |\hat{V}_+\rangle, |\hat{V}_-\rangle, |\hat{U}_+\rangle, |\hat{U}_-\rangle\}$$

and draw the weight (root) diagram.

2.2) In the adjoint representation, show that

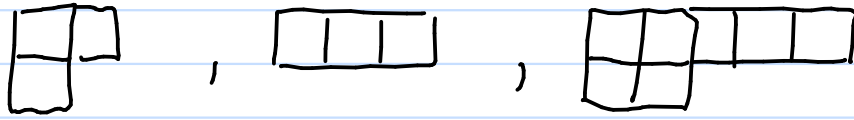
$$2.2.1) \hat{U}_+ |\hat{T}_+\rangle \propto |\hat{V}_+\rangle$$

$$2.2.2) \hat{V}_- |\hat{T}_+\rangle \propto |\hat{U}_-\rangle$$

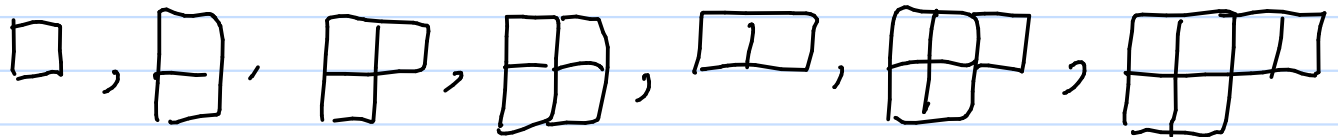
and draw the action on the weight diagram

3). Find the angle between 2 simple roots in the $SU(3)$

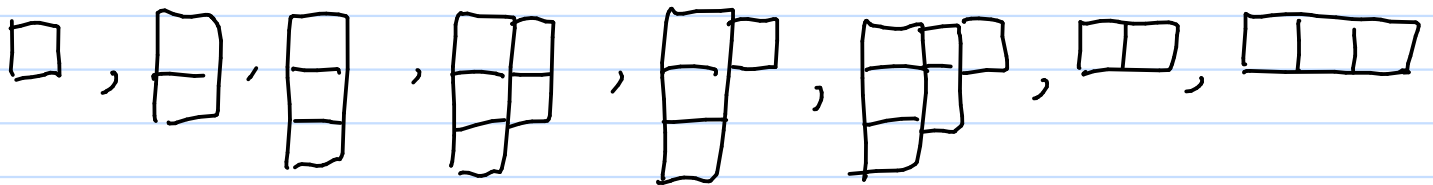
4). Find the Dynkin coefficients for the following Young diagrams in $SU(3)$



5). Find the dimension of the representation of each Young diagrams in $SU(3)$



6). Find the dimension of the representation of each Young diagrams in $SU(4)$



7). Find the tensor decomposition of $SU(2)$ reps and work out their dimensions :

7.1) $\square \otimes \square$, 7.2) $\square \otimes \square$

7.3) $\square \otimes \square$, 7.2) $\square \otimes \square$

8). Find the tensor decomposition of $SU(3)$ reps and work out their dimensions :

$$8.1) \quad \square \otimes \square$$

$$8.2) \quad \square \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$$

$$8.3) \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$$

$$8.4) \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \otimes \square$$

$$8.5) \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$$

$$8.6) \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \otimes \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$